

Lecture 20

Secret Sharing Schemes: Shamir, CRT, and Brickell

SEC 595: Encrypted Computing

Roadmap

Lecture 19 defined access structures and the two requirements on a secret sharing scheme: correctness for authorized sets and privacy for unauthorized ones. This lecture gives three schemes. Section 1 presents Shamir's threshold scheme, built on Lagrange interpolation (Lecture 2). Section 2 presents the CRT-based threshold scheme of Mignotte, built on the CRT (Lecture 1), and shows that it is not perfectly private. Section 3 presents Brickell's scheme for a general access structure. Section 4 compares the three.

1. Shamir's Scheme

1.1 Interpolation

Shamir's scheme is a (t, n) -threshold scheme. It rests on one fact from Lecture 2.

Proposition 1 (Lagrange interpolation). *Let p be prime. Given t points $(x_1, y_1), \dots, (x_t, y_t)$ in $\mathbb{F}_p^2$ with distinct x_j , there is exactly one polynomial q of degree at most $t - 1$ with $q(x_j) = y_j$ for all j .*

Existence is the Lagrange formula; uniqueness holds because the difference of two such polynomials has degree at most $t - 1$ and t roots, so it is zero. The second step needs a field, which is why p is prime.

The idea of the scheme: the secret is the constant term of a polynomial, the shares are its values, and reconstruction interpolates back to the constant term.

1.2 Dealing

Definition 1 (Shamir's scheme). Let p be a prime with $p > n$ and let $S \in \mathbb{F}_p$ be the secret.

- *Dealing.* Choose $a_1, \dots, a_{t-1}$ uniformly in $[0, p)$ and set

$$q(x) = a_0 + a_1x + \dots + a_{t-1}x^{t-1} \bmod p, \quad a_0 = S.$$

Participant P_i receives the pair (S_i, i) with $S_i = q(i) \bmod p$, for $i = 1, \dots, n$.

- *Reconstruction.* Any t participants with pairs (y_j, x_j) compute

$$S = q(0) = \sum_{j=1}^t y_j L_j \bmod p, \quad L_j = \prod_{m \neq j} \frac{0 - x_m}{x_j - x_m}.$$

The $t - 1$ random coefficients are the dealer's random string, one fewer than the threshold. The node $x = 0$ is never handed out, since $q(0) = S$. The coefficient $L_j = \ell_j(0)$ is the Lagrange basis polynomial evaluated at 0; it depends only on the nodes, that is, on *who* turned up, and can be computed before any share is revealed.

Example 1 (Distribution: $S = 13$, $n = 5$, $t = 3$, $p = 17$). With random coefficients $a_1 = 10$, $a_2 = 2$,

$$q(x) = 13 + 10x + 2x^2 \pmod{17}.$$

The shares are

$$\begin{aligned} S_1 &= 13 + 10 + 2 = 25 \equiv 8, & S_2 &= 13 + 20 + 8 = 41 \equiv 7, & S_3 &= 13 + 30 + 18 = 61 \equiv 10, \\ S_4 &= 13 + 40 + 32 = 85 \equiv 0, & S_5 &= 13 + 50 + 50 = 113 \equiv 11. \end{aligned}$$

$P_1, \dots, P_5$ receive $(8, 1)$, $(7, 2)$, $(10, 3)$, $(0, 4)$, $(11, 5)$, each written as (share, node). The dealer then discards a_1, a_2 .

Example 2 (Recovery from two node sets). (a) *Nodes 2, 3, 4*, pooled pairs $(7, 2)$, $(10, 3)$, $(0, 4)$:

$$\begin{aligned} L_1 &= \frac{-3}{2-3} \cdot \frac{-4}{2-4} = 3 \cdot 2 = 6, \\ L_2 &= \frac{-2}{3-2} \cdot \frac{-4}{3-4} = (-2) \cdot 4 = -8 \equiv 9, \\ L_3 &= \frac{-2}{4-2} \cdot \frac{-3}{4-3} = (-1) \cdot (-3) = 3. \end{aligned}$$

Then $S = 7 \cdot 6 + 10 \cdot (-8) + 0 \cdot 3 = 42 - 80 = -38 \equiv 13 \pmod{17}$. ✓

(b) *Nodes 1, 4, 5*, pooled pairs $(8, 1)$, $(0, 4)$, $(11, 5)$. Here the divisions are not exact, so we invert modulo 17; note $3^{-1} \equiv 6$ since $3 \cdot 6 = 18 \equiv 1$:

$$\begin{aligned} L_1 &= \frac{-4}{1-4} \cdot \frac{-5}{1-5} = \frac{20}{12} = \frac{5}{3} \equiv 5 \cdot 6 = 30 \equiv 13, \\ L_2 &= \frac{-1}{4-1} \cdot \frac{-5}{4-5} = \frac{5}{-3} \equiv -5 \cdot 6 = -30 \equiv 4, \\ L_3 &= \frac{-1}{5-1} \cdot \frac{-4}{5-4} = \frac{4}{4} = 1. \end{aligned}$$

Then $S = 8 \cdot 13 + 0 \cdot 4 + 11 \cdot 1 = 104 + 11 = 115 \equiv 13 \pmod{17}$, since $115 = 6 \cdot 17 + 13$. ✓

(c) *Why they agree.* Both triples of points lie on the same polynomial of degree 2, and by uniqueness there is only one such polynomial through either triple. Its constant term is 13. The coefficients L_j differ because they belong to the node set, not to the secret.

1.3 Correctness and Privacy

Proposition 2 (Correctness). *Any t participants recover S with probability 1.*

Proof. q has degree at most $t - 1$ and passes through their t points, whose nodes are distinct since $n < p$. By uniqueness the Lagrange formula reconstructs q , and $\sum_j y_j L_j = q(0) = S$. □

Proposition 3 (Perfect privacy). *Let T be any set of $t - 1$ participants. For every candidate secret s and every value of their shares, exactly one choice of $(a_1, \dots, a_{t-1})$ is consistent. Hence every secret is equally likely given T 's shares.*

Proof. The consistent polynomials are those through $(0, s)$ and the $t - 1$ points held by T . These t points have distinct nodes, because no participant holds node 0, so exactly one polynomial of degree at most $t - 1$ passes through them. Since the coefficients are uniform, each secret occurs with the same probability $p^{-(t-1)}$. $\square$

All p secrets remain consistent with T 's view, each in exactly one way, so privacy holds exactly rather than approximately. A set of fewer than $t - 1$ participants knows even less. Each share is a single element of $\mathbb{F}_p$, the same size as the secret.

2. CRT-Based Secret Sharing

2.1 The Scheme

Replace “evaluate a polynomial at n points” by “reduce an integer modulo n numbers”. Reconstruction is then the CRT (Lecture 1).

Definition 2 (Mignotte’s scheme). Choose public, pairwise coprime moduli $d_1 < d_2 < \dots < d_n$ and a secret S satisfying the **size condition**

$$\underbrace{d_1 d_2 \cdots d_t}_{t \text{ smallest}} > S > \underbrace{d_{n-t+2} \cdots d_n}_{t-1 \text{ largest}}.$$

Participant P_i receives (S_i, d_i) with $S_i = S \bmod d_i$. Any t participants solve the system $x \equiv S_i \pmod{d_i}$ by the CRT.

The two inequalities do different jobs.

- *Left: correctness.* Any t moduli have product at least $d_1 \cdots d_t > S$, so the CRT solution below that product is S .
- *Right: the threshold.* Any $t - 1$ moduli have product at most $d_{n-t+2} \cdots d_n < S$, so they determine S only modulo a number smaller than S , leaving more than one candidate.

The threshold is enforced entirely by the sizes of the moduli.

Proposition 4 (Correctness). *Any t participants B recover S .*

Proof. The CRT returns the unique $x \in [0, M_B)$, $M_B = \prod_{i \in B} d_i$, with $x \equiv S_i \pmod{d_i}$ for $i \in B$. Since $0 \leq S < d_1 \cdots d_t \leq M_B$, $x = S$. $\square$

2.2 Worked Instance

Example 3 ($S = 117$, $n = 5$, $t = 3$). (a) *Size condition.* Take $d = 4, 5, 7, 9, 11$. They are pairwise coprime (4 and 9 are not prime, but that is not required). Then

$$d_1 d_2 d_3 = 4 \cdot 5 \cdot 7 = 140 > 117 > 99 = 9 \cdot 11 = d_4 d_5.$$

(b) *Shares.*

$$117 \bmod 4 = 1, \quad 117 \bmod 5 = 2, \quad 117 \bmod 7 = 5, \quad 117 \bmod 9 = 0, \quad 117 \bmod 11 = 7.$$

The pairs are $(1, 4)$, $(2, 5)$, $(5, 7)$, $(0, 9)$, $(7, 11)$.

(c) *Recovery* from $(1, 4), (2, 5), (7, 11)$. Let $M = 4 \cdot 5 \cdot 11 = 220$, $M_i = M/d_i$, and $e_i = M_i (M_i^{-1} \bmod d_i)$:

$$\begin{aligned} M_1 = 55 &\equiv 3 \pmod{4}, & 3^{-1} &\equiv 3, & e_1 &= 55 \cdot 3 = 165, \\ M_2 = 44 &\equiv 4 \pmod{5}, & 4^{-1} &\equiv 4, & e_2 &= 44 \cdot 4 = 176, \\ M_3 = 20 &\equiv 9 \pmod{11}, & 9^{-1} &\equiv 5, & e_3 &= 20 \cdot 5 = 100. \end{aligned}$$

(Check: $9 \cdot 5 = 45 = 4 \cdot 11 + 1$.) Then

$$S = (1 \cdot 165 + 2 \cdot 176 + 7 \cdot 100) \bmod 220 = 1217 \bmod 220 = 117. \checkmark$$

The answer is unique because the CRT gives a unique residue modulo 220, and $117 < 140 \leq 220$.

(d) $n = 7, t = 5$. The condition reads $d_1 \cdots d_5 > S > d_6 d_7$. The pairwise coprime moduli 11, 13, 17, 19, 23, 25, 27 give

$$11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 = 1062347 > S > 675 = 25 \cdot 27,$$

so any S with $675 < S < 1062347$ will do.

2.3 How Private Is It?

The shares are a deterministic function of S : the scheme has no random string. The right inequality guarantees only that $t - 1$ shares leave *more than one* candidate, not that all candidates are equally likely. So Mignotte's scheme is **not** perfectly private in the sense of Lecture 19.

Example 4 (Two participants recover the secret). In Example 3, the unauthorized pair $\{P_4, P_5\}$ holds $S \equiv 0 \pmod{9}$ and $S \equiv 7 \pmod{11}$. By the CRT, $S \equiv 18 \pmod{99}$. Below 140 the candidates are 18 and 117. The moduli are public, so the pair also knows $S > 99$, which excludes 18: they recover $S = 117$.

Definition 3 (Asmuth–Bloom). Add a public modulus m_0 , coprime to the d_i , with

$$m_0 d_{n-t+2} \cdots d_n < d_1 \cdots d_t.$$

For a secret $0 \leq S < m_0$, draw a random A with $y = S + Am_0 < d_1 \cdots d_t$, share y as above, and recover $S = y \bmod m_0$.

The random A restores the missing randomness: $t - 1$ shares fix y only modulo a number $P < d_1 \cdots d_t / m_0$ coprime to m_0 , so the values of y consistent with them include m_0 consecutive terms of a progression with difference P , which cover every residue modulo m_0 . This fix cannot be applied with the moduli of Example 3: it would need $99m_0 < 140$, and no integer $m_0 \geq 2$ satisfies that.

3. Brickell's Scheme

3.1 The Scheme

Shamir and CRT realise threshold structures. Brickell's scheme realises a *general* access structure $\mathcal{A}$ (Lecture 19).

Definition 4 (Brickell's scheme). Fix a prime p and a dimension d . Give each participant a public vector $V_i \in \mathbb{F}_p^d$, chosen so that a set B is authorized exactly when $(1, 0, \dots, 0) \in \text{span}\{V_i : i \in B\}$.

- *Dealing.* Let $a = (a_1, \dots, a_d)$ with $a_1 = S$ and $a_2, \dots, a_d$ uniform in $[0, p)$. P_i receives (S_i, i) with $S_i = a \cdot V_i \bmod p$.
- *Reconstruction.* An authorized set B finds scalars c_i with $\sum_{i \in B} c_i V_i = (1, 0, \dots, 0)$ and outputs the value $\sum_{i \in B} c_i S_i \bmod p$.

As in Shamir, the secret sits in one distinguished slot and randomness fills the other $d - 1$. The c_i come from a linear system in the public vectors alone, so, like the L_j , they can be computed before any share is revealed. Constructing the V_i is the whole difficulty: every authorized set must span the target and every unauthorized set must not.

Proposition 5 (Correctness). *If $\sum_{i \in B} c_i V_i = (1, 0, \dots, 0)$, then $\sum_{i \in B} c_i S_i \equiv S \pmod{p}$.*

Proof. $\sum_i c_i (a \cdot V_i) = a \cdot (\sum_i c_i V_i) = a \cdot (1, 0, \dots, 0) = a_1 = S$. □

Proposition 6 (Privacy). *If $(1, 0, \dots, 0) \notin \text{span}\{V_i : i \in T\}$, the shares of T have the same distribution for every secret.*

Proof sketch. Since the target lies outside the span, there is w with $w \cdot V_i = 0$ for all $i \in T$ and $w \cdot (1, 0, \dots, 0) = 1$. The bijection $a \mapsto a + (S' - S)w$ changes the secret from S to S' and leaves every share of T unchanged, so it carries one distribution onto the other. □

3.2 Worked Instance

Example 5 ($S = 99$, $n = 4$, $d = 3$, $p = 127$). *Step 1: parameters.* $S = 99$, $n = 4$ participants, dimension $d = 3$, prime $p = 127$.

Step 2: access structure. The minimal authorized sets are $\{P_1, P_2, P_3\}$ and $\{P_1, P_4\}$. No threshold structure gives this: $\{P_1, P_4\}$ is authorized and $\{P_2, P_3\}$ is not, though both have size two.

Step 3: vectors.

$$V_1 = (0, 1, 0), \quad V_2 = (1, 0, 1), \quad V_3 = (0, 1, -1), \quad V_4 = (1, 1, 0),$$

which realise the two minimal sets:

$$V_2 + V_3 - V_1 = (1, 0, 0), \quad V_4 - V_1 = (1, 0, 0).$$

Step 4: secret vector. With random $a_2 = 55$, $a_3 = 38$, $a = (99, 55, 38)$.

Step 5: shares.

$$\begin{aligned} S_1 &= a \cdot V_1 = 55, \\ S_2 &= a \cdot V_2 = 99 + 38 = 137 \equiv 10, \\ S_3 &= a \cdot V_3 = 55 - 38 = 17, \\ S_4 &= a \cdot V_4 = 99 + 55 = 154 \equiv 27 \pmod{127}. \end{aligned}$$

Step 6: recovery. For $\{P_1, P_2, P_3\}$, $c = (-1, 1, 1)$:

$$-55 + 10 + 17 = -28 \equiv 99 \pmod{127}. \quad \checkmark$$

For $\{P_1, P_4\}$, $c = (-1, 1)$:

$$-55 + 27 = -28 \equiv 99 \pmod{127}. \quad \checkmark$$

Step 7: an unauthorized set. A combination of V_2, V_3 is

$$c_2(1, 0, 1) + c_3(0, 1, -1) = (c_2, c_3, c_2 - c_3).$$

Matching $(1, 0, 0)$ forces $c_2 = 1$ and $c_3 = 0$, but then the third coordinate is $1 \neq 0$. So the target is not in the span, and by the privacy proposition $\{P_2, P_3\}$ learns nothing about S . Here $w = (1, -1, -1)$ satisfies $w \cdot V_2 = w \cdot V_3 = 0$ and $w \cdot (1, 0, 0) = 1$.

4. Security and Comparison

Definition 5 (Information-theoretic security). A scheme is information-theoretically secure if its guarantee holds against an adversary with unlimited computing power.

Shamir's privacy argument is a statement about degrees: it names no adversary, no running time and no hardness assumption. Brickell's is likewise a statement about spans. By contrast, the encryption schemes of this course are computational: Regev and GSW rest on LWE, ElGamal on DDH, Paillier on composite residuosity. Mignotte's scheme needs no assumption either, but it leaks; Asmuth–Bloom repairs this.

	Shamir	CRT	Brickell
Structure	(t, n)	(t, n)	general $\mathcal{A}$
Secret in	$a_0 = q(0)$	S	a_1
Share	$q(i)$	$S \bmod d_i$	$a \cdot V_i$
Recovery	Lagrange at 0	CRT	solve for c_i
Privacy	exact	interval-dependent	exact

In all three, reconstruction is a fixed linear function of the shares, determined by which participants turned up.

Summary

1. Shamir: a polynomial q of degree at most $t - 1$ with $q(0) = S$; shares $(q(i), i)$; recovery is Lagrange interpolation at $x = 0$.
2. Any $t - 1$ Shamir shares are consistent with every secret in exactly one way, so privacy is exact and information-theoretic.
3. CRT (Mignotte): shares $S \bmod d_i$, threshold from $d_1 \cdots d_t > S > d_{n-t+2} \cdots d_n$. The shares are deterministic and can leak the secret; Asmuth–Bloom adds randomness.
4. Brickell: public vectors V_i , shares $a \cdot V_i \bmod p$; authorized sets combine their vectors to $(1, 0, \dots, 0)$, which selects $a_1 = S$.
5. All three put the secret in one slot and randomness, if any, in the rest; reconstruction is linear in the shares.

Exercises

1. Recover the secret of Example 1 from the shares at nodes 1, 2, 3. Compute the Lagrange coefficients first.
2. Set up a $(2, 3)$ Shamir scheme over $\mathbb{F}_{11}$ with $S = 4$ and $a_1 = 7$. Compute the three shares and recover S from each pair.

3. Using the shares $(7, 2)$ and $(10, 3)$ of Example 1, find the quadratic over $\mathbb{F}_{17}$ through $(0, 0)$, $(2, 7)$, $(3, 10)$. Which part of Proposition 3 does it illustrate?
4. Redo the recovery in Example 3 from P_3, P_4, P_5 , with moduli 7, 9, 11 and shares 5, 0, 7. Verify that the idempotents sum to 1 modulo M .
5. For the pair $\{P_1, P_2\}$ of Example 3, list the candidates below 140 and those with $S > 99$. Compare with Example 4.
6. In Example 5, find c_1, c_2, c_4 for the authorized set $\{P_1, P_2, P_4\}$ and check that they recover 99.
7. Show that $(1, 0, 0)$ is not in the span of V_2, V_3, V_4 . What does this say about P_1 ?